FIBRE ISOMORPHISM SPREADS TO A NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Finite in a neighbourhood. coherent-lemma-proper-finite-fibre-finite-in-neighbourhood (cross-chapter, named). Makes p finite near the fibre.
- Flatness criterion. more-morphisms-theorem-criterion-flatness-fibre-Noetherian (cross-chapter, named). Makes p flat near the fibre.

Source reference. Original reference 02KG: focused-seeds/geometry/S9-eqv-isomorphism-in-neighbourhood/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02OH: focused-seeds/geometry/S9-eqv-isomorphism-in-neighbourhood/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01B8: focused-seeds/geometry/S9-eqv-isomorphism-in-neighbourhood/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 039B: focused-seeds/geometry/S9-eqv-isomorphism-in-neighbourhood/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01W5: focused-seeds/geometry/S9-eqv-isomorphism-in-neighbourhood/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01UA: focused-seeds/geometry/S9-eqv-isomorphism-in-neighbourhood/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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Source reference. Original reference 04PV: focused-seeds/geometry/S9-eqv-isomorphism-in-neighbourhood/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a Noetherian scheme. Let $s \in S$. Let $p : X \to Y$ be a morphism of schemes over S. Assume*

(1) *$Y \to S$ and $X \to S$ proper,*
(2) *X is flat over S,*
(3) *$X_s \to Y_s$ an isomorphism.*

Then there exists an open neighbourhood $U \subset S$ of s such that the base change $X_U \to Y_U$ is an isomorphism.

Proof. The morphism p is proper by Morphisms, Lemma 01W5. By Cohomology of Schemes, Lemma 02OH there is an open $Y_s \subset V \subset Y$ such that $p|_{p^{-1}(V)} : p^{-1}(V) \to V$ is finite. By More on Morphisms, Theorem 039B there is an open $X_s \subset U \subset X$ such that $p|_U : U \to Y$ is flat. After removing the images of $X \setminus U$ and $Y \setminus V$ (which are closed subsets not containing s) we may assume p is flat and finite. Then p is open (Morphisms, Lemma 01UA) and $Y_s \subset p(X) \subset Y$ hence after shrinking S we may assume p is surjective. As $p_s : X_s \to Y_s$ is an isomorphism, the map

$$p^\sharp : \mathcal{O}_Y \longrightarrow p_*\mathcal{O}_X$$

of coherent $\mathcal{O}_Y$-modules (p is finite) becomes an isomorphism after pullback by $i : Y_s \to Y$ (by Cohomology of Schemes, Lemma 02KG for example). By Nakayama's lemma, this implies that $\mathcal{O}_{Y,y} \to (p_*\mathcal{O}_X)_y$ is surjective for all $y \in Y_s$. Hence there is an open $Y_s \subset V \subset Y$ such that $p^\sharp|_V$ is surjective (Modules, Lemma 01B8). Hence after shrinking S once more we may assume $p^\sharp$ is surjective which means that p is a closed immersion (as p is already finite). Thus now p is a surjective flat closed immersion of Noetherian schemes and hence an isomorphism, see Morphisms, Section 04PV. □